T05

Seminar In-Class Activity Deck

2026 AI-First · CS423/CSC15003

FIT @ HCMUS · Department of Software Engineering

Performance Testing

JMeter is still king — but AI now writes realistic load scenarios from your logs.

ACTIVITY: Workload Model Bake-off

Duration: ~25 minutes · For audience teams (3–4 SV)

Why this matters

Performance testing on EShop combines load, stress, and spike profiles against the Node.js + SQLite backend. You will design a realistic mixed-workload model — search-heavy, checkout-bursty — and run it from JMeter or k6. The AI angle: use ChatGPT/Claude to translate your assumptions or an HTTP log into a k6 scenario, then audit the script for unrealistic distributions.

Ground Rules for Today's Activity

- Work in your team of 3–4. One designated note-taker per team.
- Read the worksheet first; ask the facilitator before guessing.
- AI tools are allowed — but every output must be cross-checked manually.
- At the end, each team submits ONE minute-paper for audience attendance credit.

What you (the audience) will leave with

Learning outcomes

- Design a workload model: % of search vs cart-add vs checkout per virtual user-minute.
- Implement the model in JMeter (.jmx) AND in k6 (JavaScript).
- Run a baseline load (50 VU) and a spike (50→500 VU in 30s).

Expected takeaways

✓ Realistic workload modelling matters more than raw VU count.
✓ AI helps shape scenarios but cannot verify business realism — humans must.
✓ SQLite is fine for HW; production SUTs would need pooled Postgres + connection limits.

Tool snapshot — traditional vs AI-augmented

Traditional

Apache JMeter
GUI + CLI; rich plugin ecosystem; XML scripts.

k6 (Grafana Labs)
JavaScript scripting, cloud-friendly, modern reports.

Locust (Python)
Pure-Python scenarios, easy distributed mode.

AI-augmented

ChatGPT / Claude for k6
Prompt with a sample HAR or log; generate k6 script; audit.

Grafana k6 AI
AI-suggested correlations and thresholds from past runs.

Akamas (AI optimisation)
Tunes JVM / DB parameters; uses ML for parameter search — discuss only, not required to run.

Activity: Workload Model Bake-off

Audience teams design a realistic workload distribution for EShop Black-Friday in 15 minutes.

What you will need on your laptop

- EShop running locally (frontend + backend) OR ability to connect to the lecturer's stage env.
- Your favourite text editor + a terminal.
- An AI tool of your choice (ChatGPT/Claude/Gemini/Copilot/Cursor — free tier OK).
- The Activity_Worksheet.md provided 3 days before this seminar.

Activity steps — keep tight to the clock

Time	Step
0:00–0:03	Facilitator shows the EShop site map + the 22 FRs.
0:03–0:12	Teams sketch %-mix per user-minute on a worksheet.
0:12–0:17	Each team posts the mix; facilitator collects on the board.
0:17–0:22	Facilitator runs the winning team's mix as a k6 dry-run; debrief.
0:22–0:25	Teams note one assumption they would have changed.

Debrief prompts — discuss in 5 minutes

Q1	What surprised you the most?
Q2	Where did the AI tool save you time vs add work?
Q3	Identify ONE failure mode you would prevent next time.
Q4	If you had 1 more hour, what would you investigate next?

Each team writes ONE answer per prompt on the minute paper (handed in at end of class).

Takeaways + further reading

3 things to remember

✓ Realistic workload modelling matters more than raw VU count.
✓ AI helps shape scenarios but cannot verify business realism — humans must.
✓ SQLite is fine for HW; production SUTs would need pooled Postgres + connection limits.

Further reading

- Molyneaux — 'The Art of Application Performance Testing' (Chs. 2, 6).
- Grafana k6 — getting-started + thresholds docs.
- Brendan Gregg — 'Systems Performance' (intro).

Submit minute-paper now · Thank you!